

2. [Maximum mark: 29]

The following question uses Maclaurin series to investigate approximations of mathematical constants and the accuracies of such approximations.

(a) Given $|x| < 1$, find the sum to infinity of the geometric series $1 - x^2 + x^4 - x^6 + \dots$. [2]

(b) Hence, use integration to show that the Maclaurin series of $\arctan x$ may be expressed as $\arctan x = x - \frac{x^3}{3} + \frac{x^5}{5} - \frac{x^7}{7} + \dots$. [3]

(c) Using $x = \frac{1}{\sqrt{3}}$ and the first **three** (non-zero) terms of the Maclaurin series of $\arctan x$, find an approximation for π to three decimal places. [3]

The Maclaurin series of $\arctan x$ is an example of an alternating series, *ie* a series where consecutive terms are positive and negative. Consider the following theorem.

Theorem: For alternating series with terms of decreasing magnitude, the error obtained in using a finite number of terms is less than or equal to the absolute value of the next term in the sequence.

Using the theorem, the maximum error in using the first three (non-zero) terms as an approximation to $\arctan x$ is given by $\left| -\frac{x^7}{7} \right|$. In other words, $\left| \arctan x - \left(x - \frac{x^3}{3} + \frac{x^5}{5} \right) \right| \leq \left| -\frac{x^7}{7} \right|$.

(d) Determine how many (non-zero) terms of the series would need to be used, such that the error in approximating $\arctan\left(\frac{1}{\sqrt{3}}\right)$ is less than 0.0001. [3]

(e) By using integration by parts, show that $\int_0^{\frac{1}{\sqrt{3}}} \arctan x \, dx = \frac{\pi}{6\sqrt{3}} - \frac{1}{2} \ln \frac{4}{3}$. [4]

(f) Determine the value of $\int_0^{\frac{1}{\sqrt{3}}} \left(x - \frac{x^3}{3} + \frac{x^5}{5} - \frac{x^7}{7} \right) dx$, giving your answer to six decimal places. [2]

(g) Use the results in parts (e) and (f) to find an approximation for π . Give your answer to four decimal places. [2]

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(Question 2 continued)

$\int_0^{\frac{1}{\sqrt{3}}} \left(x - \frac{x^3}{3} + \frac{x^5}{5} - \frac{x^7}{7} + \dots \right) dx$ may be considered as the sum of alternating terms.

Hence, $\int_0^{\frac{1}{\sqrt{3}}} \left(x - \frac{x^3}{3} + \frac{x^5}{5} - \frac{x^7}{7} + \dots \right) dx = \int_0^{\frac{1}{\sqrt{3}}} x \, dx + \int_0^{\frac{1}{\sqrt{3}}} \left(-\frac{x^3}{3} \right) dx + \int_0^{\frac{1}{\sqrt{3}}} \left(\frac{x^5}{5} \right) dx + \int_0^{\frac{1}{\sqrt{3}}} \left(-\frac{x^7}{7} \right) dx + \dots$

$\int_0^{\frac{1}{\sqrt{3}}} \arctan x \, dx$ is approximated using the sum of the first four definite integrals.

(h) Verify that the theorem used in part (d) holds in this case. [5]

Suppose that the maximum error in approximating $\int_0^{\frac{1}{\sqrt{3}}} \arctan x \, dx$ is required to be at most 1×10^{-6} .

(i) Determine the smallest number of (non-zero) terms of the Maclaurin series for $\arctan x$ that should be used. [5]
